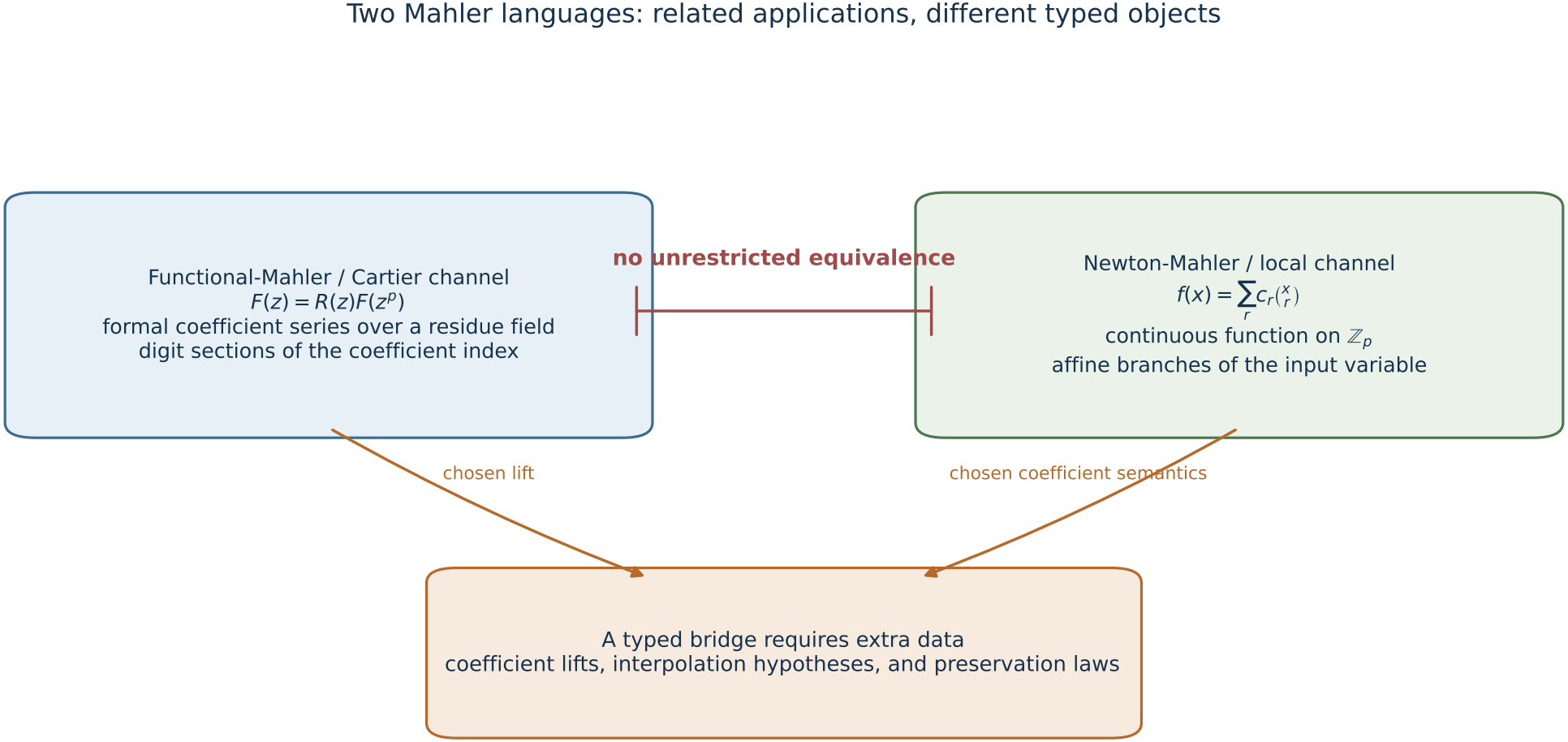
Two Mahler languages: related applications, different typed objects
Functional-Mahler / Cartier channel
$F(z) = R(z)F(z^p)$
formal coefficient series over a residue field
digit sections of the coefficient index
no unrestricted equivalence
Newton-Mahler / local channel
$f(x) = \sum_r c_r \binom{x}{r}$
continuous function on $\mathbb{Z}_p$
affine branches of the input variable
chosen lift
chosen coefficient semantics
A typed bridge requires extra data
coefficient lifts, interpolation hypotheses, and preservation laws
Equal terminology does not identify categories, variables, coefficient rings, or normalization targets.